TASK 9: SALES AND CUSTOMER ANALYTICS SYSTEM

PROJECT DETAILS

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SCENARIO

The management of **PALETTE: Pixel & Pages** wants to analyze sales data to understand customer purchasing behavior, product performance (Books), and revenue trends. Management requires reports to identify top customers, best-selling products (Books), and profitable categories (Book Categories).

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1. APPLY SQL AGGREGATE FUNCTIONS

Aggregate functions perform calculations on multiple records and return a single summarized value.

Function	Purpose	Bookstore Example
COUNT()	Counts number of records	Total Orders
SUM()	Calculates total value	Total Sales Amount
AVG()	Finds average value	Average Order Value
MIN()	Finds minimum value	Lowest Book Price
MAX()	Finds maximum value	Highest Book Price

---

2. GENERATE TOTAL SALES REPORTS

2.1 Total Number of Orders Placed

```
SQL> SELECT COUNT(*) AS TOTAL_ORDERS
  2  FROM ORDERS;

TOTAL_ORDERS
------------
          20
```

2.2 Total Revenue Generated from Sales

```
SQL> SELECT SUM(TOTAL_AMOUNT) AS TOTAL_REVENUE
  2  FROM ORDERS;

TOTAL_REVENUE
-------------
        14737
```

2.3 Average Order Value

```
SQL> SELECT ROUND(AVG(TOTAL_AMOUNT), 2) AS AVERAGE_ORDER_VALUE
  2  FROM ORDERS;

AVERAGE_ORDER_VALUE
-------------------
             736.85
```

2.4 Highest and Lowest Order Amount

```
SQL> SELECT MAX(TOTAL_AMOUNT) AS HIGHEST_ORDER,
  2  MIN(TOTAL_AMOUNT) AS LOWEST_ORDER
  3  FROM ORDERS;

HIGHEST_ORDER LOWEST_ORDER
------------- ------------
         1250          399
```

2.5 Total Sales for a Specific Period

```
SQL> SELECT SUM(TOTAL_AMOUNT) AS TOTAL_REVENUE
  2  FROM ORDERS
  3  WHERE ORDER_DATE BETWEEN TO_DATE('15-MAR-26','DD-MON-RR')
  4  AND TO_DATE('30-MAR-26','DD-MON-RR');

TOTAL_REVENUE
-------------
        10641
```

---

3. CUSTOMER PURCHASE ANALYSIS

3.1 Total Number of Orders Placed by Each Customer

```
SQL> SELECT C.CUSTOMER_ID,
  2  C.FIRST_NAME,
  3  COUNT(O.ORDER_ID) AS TOTAL_ORDERS
  4  FROM CUSTOMER C
  5  LEFT JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID,
  8  C.FIRST_NAME
  9  ORDER BY TOTAL_ORDERS DESC;

CUSTOMER_ID          FIRST_NAME           TOTAL_ORDERS
-------------------- -------------------- ------------
CUST-101             Ananya                          2
CUST-107             Kabir                           1
CUST-110             Meera                           1
CUST-113             Nikhil                          1
CUST-114             Pooja                           1
CUST-112             Anika                           1
CUST-116             Sneha                           1
CUST-119             Rahul                           1
CUST-102             Rohan                           1
CUST-103             Priya                           1
CUST-115             Vikram                          1

CUSTOMER_ID          FIRST_NAME           TOTAL_ORDERS
-------------------- -------------------- ------------
CUST-118             Kavya                           1
CUST-117             Aditya                          1
CUST-105             Aarav                           1
CUST-106             Ishita                          1
CUST-108             Diya                            1
CUST-120             Neha                            1
CUST-109             Arjun                           1
CUST-111             Rohan                           1
CUST-104             Vikram                          0

20 rows selected.
```

3.2 Total Amount Spent by Each Customer

```
SQL> SELECT C.CUSTOMER_ID,
  2  C.FIRST_NAME,
  3  NVL(SUM(O.TOTAL_AMOUNT),0) AS TOTAL_SPENT
  4  FROM CUSTOMER C
  5  LEFT JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID,
  8  C.FIRST_NAME
  9  ORDER BY TOTAL_SPENT DESC;

CUSTOMER_ID          FIRST_NAME           TOTAL_SPENT
-------------------- -------------------- -----------
CUST-101             Ananya                      2348
CUST-103             Priya                       1149
CUST-111             Rohan                        999
CUST-115             Vikram                       899
CUST-112             Anika                        850
CUST-106             Ishita                       799
CUST-114             Pooja                        799
CUST-118             Kavya                        750
CUST-117             Aditya                       699
CUST-105             Aarav                        699
CUST-110             Meera                        650

CUSTOMER_ID          FIRST_NAME           TOTAL_SPENT
-------------------- -------------------- -----------
CUST-108             Diya                         599
CUST-102             Rohan                        599
CUST-120             Neha                         550
CUST-113             Nikhil                       550
CUST-107             Kabir                        499
CUST-109             Arjun                        450
CUST-116             Sneha                        450
CUST-119             Rahul                        399
CUST-104             Vikram                         0

20 rows selected.
```

3.3 Average Spending Per Customer

```
SQL> SELECT C.CUSTOMER_ID,
  2  C.FIRST_NAME,
  3  ROUND(NVL(AVG(O.TOTAL_AMOUNT),0),2) AS AVERAGE_SPENDING
  4  FROM CUSTOMER C
  5  LEFT JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID,
  8  C.FIRST_NAME
  9  ORDER BY AVERAGE_SPENDING DESC;

CUSTOMER_ID          FIRST_NAME           AVERAGE_SPENDING
-------------------- -------------------- ----------------
CUST-101             Ananya                           1174
CUST-103             Priya                            1149
CUST-111             Rohan                             999
CUST-115             Vikram                            899
CUST-112             Anika                             850
CUST-106             Ishita                            799
CUST-114             Pooja                             799
CUST-118             Kavya                             750
CUST-117             Aditya                            699
CUST-105             Aarav                             699
CUST-110             Meera                             650

CUSTOMER_ID          FIRST_NAME           AVERAGE_SPENDING
-------------------- -------------------- ----------------
CUST-108             Diya                              599
CUST-102             Rohan                             599
CUST-120             Neha                              550
CUST-113             Nikhil                            550
CUST-107             Kabir                             499
CUST-109             Arjun                             450
CUST-116             Sneha                             450
CUST-119             Rahul                             399
CUST-104             Vikram                              0

20 rows selected.
```

3.4 Customers with Maximum Purchase Amount

```
SQL> SELECT C.CUSTOMER_ID,
  2  C.FIRST_NAME,
  3  SUM(O.TOTAL_AMOUNT) AS TOTAL_PURCHASE
  4  FROM CUSTOMER C
  5  JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID,
  8  C.FIRST_NAME
  9  HAVING SUM(O.TOTAL_AMOUNT) =
 10  (
 11  SELECT MAX(TOTAL_SPENDING)
 12  FROM
 13  (
 14  SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SPENDING
 15  FROM ORDERS
 16  GROUP BY CUSTOMER_ID
 17  )
 18  );

CUSTOMER_ID          FIRST_NAME           TOTAL_PURCHASE
-------------------- -------------------- --------------
CUST-101             Ananya                         2348
```

3.5 Customers with Fewer Purchases

```
SQL> SELECT C.CUSTOMER_ID,
  2  C.FIRST_NAME,
  3  COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS
  4  FROM CUSTOMER C
  5  LEFT JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID,
  8  C.FIRST_NAME
  9  HAVING COUNT(O.ORDER_ID) <= 1
 10  ORDER BY NUMBER_OF_ORDERS;

CUSTOMER_ID          FIRST_NAME           NUMBER_OF_ORDERS
-------------------- -------------------- ----------------
CUST-104             Vikram                              0
CUST-117             Aditya                              1
CUST-110             Meera                               1
CUST-113             Nikhil                              1
CUST-114             Pooja                               1
CUST-112             Anika                               1
CUST-116             Sneha                               1
CUST-119             Rahul                               1
CUST-102             Rohan                               1
CUST-103             Priya                               1
CUST-115             Vikram                              1

CUSTOMER_ID          FIRST_NAME           NUMBER_OF_ORDERS
-------------------- -------------------- ----------------
CUST-118             Kavya                               1
CUST-106             Ishita                              1
CUST-108             Diya                                1
CUST-120             Neha                                1
CUST-109             Arjun                               1
CUST-111             Rohan                               1
CUST-105             Aarav                               1
CUST-107             Kabir                               1

19 rows selected.
```

---

4. FIND TOP CUSTOMERS BASED ON PURCHASE AMOUNT

4.1 Top 5 Customers Based on Total Spending

```
SQL> SELECT *
  2  FROM
  3  (
  4  SELECT C.CUSTOMER_ID,
  5  C.FIRST_NAME,
  6  SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING
  7  FROM CUSTOMER C
  8  JOIN ORDERS O
  9  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 10  GROUP BY C.CUSTOMER_ID,
 11  C.FIRST_NAME
 12  ORDER BY TOTAL_SPENDING DESC
 13  )
 14  WHERE ROWNUM <= 5;

CUSTOMER_ID          FIRST_NAME           TOTAL_SPENDING
-------------------- -------------------- --------------
CUST-101             Ananya                         2348
CUST-103             Priya                          1149
CUST-111             Rohan                           999
CUST-115             Vikram                          899
CUST-112             Anika                           850
```

4.2 Customers with Maximum Number of Orders

```
SQL> SELECT C.CUSTOMER_ID,
  2  C.FIRST_NAME,
  3  COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS
  4  FROM CUSTOMER C
  5  JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID,
  8  C.FIRST_NAME
  9  ORDER BY NUMBER_OF_ORDERS DESC;

CUSTOMER_ID          FIRST_NAME           NUMBER_OF_ORDERS
-------------------- -------------------- ----------------
CUST-101             Ananya                              2
CUST-107             Kabir                               1
CUST-110             Meera                               1
CUST-113             Nikhil                              1
CUST-114             Pooja                               1
CUST-112             Anika                               1
CUST-116             Sneha                               1
CUST-119             Rahul                               1
CUST-102             Rohan                               1
CUST-103             Priya                               1
CUST-115             Vikram                              1

CUSTOMER_ID          FIRST_NAME           NUMBER_OF_ORDERS
-------------------- -------------------- ----------------
CUST-118             Kavya                               1
CUST-106             Ishita                              1
CUST-108             Diya                                1
CUST-120             Neha                                1
CUST-109             Arjun                               1
CUST-111             Rohan                               1
CUST-117             Aditya                              1
CUST-105             Aarav                               1

19 rows selected.
```

4.3 Frequent Customers

```
SQL> SELECT C.CUSTOMER_ID,
  2  C.FIRST_NAME,
  3  COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS
  4  FROM CUSTOMER C
  5  JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID,
  8  C.FIRST_NAME
  9  HAVING COUNT(O.ORDER_ID) >= 2;

CUSTOMER_ID          FIRST_NAME           NUMBER_OF_ORDERS
-------------------- -------------------- ----------------
CUST-101             Ananya                              2
```

4.4 High-Value Customers

```
SQL> SELECT C.CUSTOMER_ID,
  2  C.FIRST_NAME,
  3  SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENT
  4  FROM CUSTOMER C
  5  JOIN ORDERS O
  6  ON C.CUSTOMER_ID = O.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID,
  8  C.FIRST_NAME
  9  HAVING SUM(O.TOTAL_AMOUNT) > 1500;

CUSTOMER_ID          FIRST_NAME           TOTAL_SPENT
-------------------- -------------------- -----------
CUST-101             Ananya                      2348
```

---

5. IDENTIFY BEST-SELLING PRODUCTS (BOOKS)

5.1 Products with Maximum Sales Quantity (Books)

```
SQL> SELECT B.BOOK_ID,
  2  B.TITLE,
  3  SUM(OD.QUANTITY) AS TOTAL_UNITS_SOLD
  4  FROM BOOK B
  5  JOIN ORDER_DETAILS OD
  6  ON B.BOOK_ID = OD.PRODUCT_ID
  7  GROUP BY B.BOOK_ID,
  8  B.TITLE
  9  ORDER BY TOTAL_UNITS_SOLD DESC;

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------
B1001
The Great Gatsby
               2

B1002
Dune
               2

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------

B1024
Angels and Demons
               1

B1004
Designing Data-Intensive Applications

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------
               1

B1022
Percy Jackson and the Lightning Thief
               1

B1005

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------
The Silent Patient
               1

B1006
Gone Girl
               1
```

```
BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------
B1007
Sherlock Holmes
                1

B1020
Around the World in 80 Days
                1

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------

B1014
The Shining
                1

B1019
Treasure Island

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------
                1

B1009
Pride and Prejudice
                1

B1011

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------
Harry Potter and the Philosopher's Stone
                1

B1008
Murder on the Orient Express
                1
```

```
BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------
B1010
Me Before You
                1

B1012
The Hobbit
                1

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------

B1003
Foundation
                1

B1018
The Diary of a Young Girl

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------
                1

B1013
Dracula
                1

B1015

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------
Wings of Fire
                1


20 rows selected.
```

5.2 Products Generating Highest Revenue (Books)

```
SQL> SELECT B.BOOK_ID,
  2  B.TITLE,
  3  SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  4  FROM BOOK B
  5  JOIN ORDER_DETAILS OD
  6  ON B.BOOK_ID = OD.PRODUCT_ID
  7  GROUP BY B.BOOK_ID,
  8  B.TITLE
  9  ORDER BY TOTAL_REVENUE DESC;

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
B1004
Designing Data-Intensive Applications
         1250

B1002
Dune
         1198

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------

B1011
Harry Potter and the Philosopher's Stone
          999

B1001
The Great Gatsby

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
          998

B1015
Wings of Fire
          899

B1012

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
The Hobbit
          850

B1006
Gone Girl
          799
```

```
BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
B1014
The Shining
          799

B1024
Angels and Demons
          750

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------

B1022
Percy Jackson and the Lightning Thief
          699

B1005
The Silent Patient

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
          699

B1003
Foundation
          650

B1010

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
Me Before You
          650

B1008
Murder on the Orient Express
          599
```

```
BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
B1020
Around the World in 80 Days
         550

B1013
Dracula
         550

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------

B1007
Sherlock Holmes
         499

B1018
The Diary of a Young Girl

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
         450

B1009
Pride and Prejudice
         450

B1019

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
Treasure Island
         399


20 rows selected.
```

5.3 Least-Selling Products (Books)

```
SQL> SELECT B.BOOK_ID,
  2  B.TITLE,
  3  SUM(OD.QUANTITY) AS TOTAL_UNITS_SOLD
  4  FROM BOOK B
  5  JOIN ORDER_DETAILS OD
  6  ON B.BOOK_ID = OD.PRODUCT_ID
  7  GROUP BY B.BOOK_ID,
  8  B.TITLE
  9  ORDER BY TOTAL_UNITS_SOLD ASC;

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------
B1013
Dracula
               1

B1018
The Diary of a Young Girl
               1

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------

B1024
Angels and Demons
               1

B1004
Designing Data-Intensive Applications

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------
               1

B1022
Percy Jackson and the Lightning Thief
               1

B1005

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------
The Silent Patient
               1

B1006
Gone Girl
               1
```

```
BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------
B1007
Sherlock Holmes
               1

B1020
Around the World in 80 Days
               1

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------

B1014
The Shining
               1

B1019
Treasure Island

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------
               1

B1015
Wings of Fire
               1

B1003

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------
Foundation
               1

B1009
Pride and Prejudice
               1
```

```
BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------
B1011
Harry Potter and the Philosopher's Stone
               1

B1008
Murder on the Orient Express
               1

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------

B1010
Me Before You
               1

B1012
The Hobbit

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------
               1

B1002
Dune
               2

B1001

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
TOTAL_UNITS_SOLD
----------------
The Great Gatsby
               2


20 rows selected.
```

5.4 Products Requiring Promotion (Books)

```
SQL> SELECT B.BOOK_ID,
  2  B.TITLE,
  3  NVL(SUM(OD.QUANTITY),0) AS QUANTITY_SOLD
  4  FROM BOOK B
  5  LEFT JOIN ORDER_DETAILS OD
  6  ON B.BOOK_ID = OD.PRODUCT_ID
  7  GROUP BY B.BOOK_ID,
  8  B.TITLE
  9  HAVING NVL(SUM(OD.QUANTITY),0) <= 1
 10  ORDER BY QUANTITY_SOLD;

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------
B1021
The Fault in Our Stars
            0

B1023
The Da Vinci Code
            0

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------

B1017
Sapiens
            0

B1016
Steve Jobs

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------
            0

B1022
Percy Jackson and the Lightning Thief
            1

B1005

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------
The Silent Patient
            1
```

```
B1006
Gone Girl
            1


BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------
B1007
Sherlock Holmes
            1

B1020
Around the World in 80 Days
            1

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------

B1014
The Shining
            1

B1019
Treasure Island

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------
            1

B1009
Pride and Prejudice
            1

B1011

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------
Harry Potter and the Philosopher's Stone
            1

B1008
Murder on the Orient Express
            1
```

```
BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------
B1010
Me Before You
            1

B1012
The Hobbit
            1

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------

B1003
Foundation
            1

B1004
Designing Data-Intensive Applications

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------
            1

B1024
Angels and Demons
            1

B1018

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------
The Diary of a Young Girl
            1

B1013
Dracula
            1
```

```
BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
QUANTITY_SOLD
-------------
B1015
Wings of Fire
            1


22 rows selected.
```

6. PERFORM CATEGORY-WISE SALES ANALYSIS (BOOK CATEGORIES)

6.1 Total Sales Generated by Each Category (Book Categories)

```
SQL> SELECT C.CATEGORY_NAME,
  2  SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  3  FROM CATEGORY C
  4  JOIN BOOK B
  5  ON C.CATEGORY_ID = B.CATEGORY_ID
  6  JOIN ORDER_DETAILS OD
  7  ON B.BOOK_ID = OD.PRODUCT_ID
  8  GROUP BY C.CATEGORY_NAME;

CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
Fiction
         998

Sci-Fi
        1848

Mystery
         499


CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
Biography
         899

Young Adult
         699

Technology
        1250


CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
Thriller
        1498

Romance
        1100

Horror
        1349


CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
Adventure
         949

Fantasy
        1849

Crime
        1349
```

```
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
History
         450


13 rows selected.
```

6.2 Number of Products Sold in Each Category (Books)

```
SQL> SELECT C.CATEGORY_NAME,
  2  SUM(OD.QUANTITY) AS PRODUCTS_SOLD
  3  FROM CATEGORY C
  4  JOIN BOOK B
  5  ON C.CATEGORY_ID = B.CATEGORY_ID
  6  JOIN ORDER_DETAILS OD
  7  ON B.BOOK_ID = OD.PRODUCT_ID
  8  GROUP BY C.CATEGORY_NAME;

CATEGORY_NAME
--------------------------------------------------------------------------------
PRODUCTS_SOLD
-------------
Fiction
            2

Sci-Fi
            3

Mystery
            1


CATEGORY_NAME
--------------------------------------------------------------------------------
PRODUCTS_SOLD
-------------
Biography
            1

Young Adult
            1

Technology
            1


CATEGORY_NAME
--------------------------------------------------------------------------------
PRODUCTS_SOLD
-------------
Thriller
            2

Romance
            2

Horror
            2


CATEGORY_NAME
--------------------------------------------------------------------------------
PRODUCTS_SOLD
-------------
Adventure
            2

Fantasy
            2

Crime
            2
```

```
CATEGORY_NAME
--------------------------------------------------------------------------------
PRODUCTS_SOLD
-------------
History
            1


13 rows selected.
```

6.3 Highest Revenue-Generating Category (Book Category)

```
SQL> SELECT *
  2  FROM
  3  (
  4  SELECT C.CATEGORY_NAME,
  5  SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  6  FROM CATEGORY C
  7  JOIN BOOK B
  8  ON C.CATEGORY_ID = B.CATEGORY_ID
  9  JOIN ORDER_DETAILS OD
 10  ON B.BOOK_ID = OD.PRODUCT_ID
 11  GROUP BY C.CATEGORY_NAME
 12  ORDER BY TOTAL_REVENUE DESC
 13  )
 14  WHERE ROWNUM = 1;

CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_REVENUE
-------------
Fantasy
         1849
```

6.4 Average Sales Per Category (Book Categories)

```
SQL> SELECT C.CATEGORY_NAME,
  2  ROUND(AVG(OD.QUANTITY * OD.PRICE),2) AS AVERAGE_SALES
  3  FROM CATEGORY C
  4  JOIN BOOK B
  5  ON C.CATEGORY_ID = B.CATEGORY_ID
  6  JOIN ORDER_DETAILS OD
  7  ON B.BOOK_ID = OD.PRODUCT_ID
  8  GROUP BY C.CATEGORY_NAME;

CATEGORY_NAME
--------------------------------------------------------------------------------
AVERAGE_SALES
-------------
Fiction
          499

Sci-Fi
          616

Mystery
          499


CATEGORY_NAME
--------------------------------------------------------------------------------
AVERAGE_SALES
-------------
Biography
          899

Young Adult
          699

Technology
         1250


CATEGORY_NAME
--------------------------------------------------------------------------------
AVERAGE_SALES
-------------
Thriller
          749

Romance
          550

Horror
        674.5


CATEGORY_NAME
--------------------------------------------------------------------------------
AVERAGE_SALES
-------------
Adventure
        474.5

Fantasy
        924.5

Crime
        674.5
```

```
CATEGORY_NAME
--------------------------------------------------------------------------------
AVERAGE_SALES
-------------
History
          450

13 rows selected.
```

---

7. GENERATE BUSINESS ANALYTICS REPORTS

7.1 Report 1: Sales Performance Report

- **Total Sales**

```
SQL> SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SALES
  2  FROM ORDERS;

TOTAL_SALES
-----------
      14737
```

- **Number of Orders**

```
SQL> SELECT COUNT(*) AS TOTAL_ORDERS
  2  FROM ORDERS;

TOTAL_ORDERS
------------
          20
```

- **Average Order Value**

```
SQL> SELECT ROUND(AVG(TOTAL_AMOUNT),2) AS AVERAGE_ORDER_VALUE
  2  FROM ORDERS;

AVERAGE_ORDER_VALUE
-------------------
             736.85
```

- **Highest Order Value**

```
SQL> SELECT MAX(TOTAL_AMOUNT) AS HIGHEST_ORDER_VALUE
  2  FROM ORDERS;

HIGHEST_ORDER_VALUE
-------------------
               1250
```

7.2 Report 2: Customer Analytics Report

- **Customer Name**
- **Number of Orders**
- **Total Spending**
- **Purchase Frequency**

```
SQL> SELECT C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
  2  COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS,
  3  SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING,
  4  CASE
  5  WHEN COUNT(O.ORDER_ID) >= 2
  6  THEN 'Frequent Customer'
  7  ELSE 'Occasional Customer'
  8  END AS PURCHASE_FREQUENCY
  9  FROM CUSTOMER C
 10  JOIN ORDERS O
 11  ON C.CUSTOMER_ID = O.CUSTOMER_ID
 12  GROUP BY C.FIRST_NAME,
 13  C.LAST_NAME;

CUSTOMER_NAME                          NUMBER_OF_ORDERS TOTAL_SPENDING
-------------------------------------- ---------------- --------------
PURCHASE_FREQUENCY
--------------------
Priya Nair                                            1           1149
Occasional Customer

Aarav Sharma                                          1            699
Occasional Customer

Arjun Nair                                            1            450
Occasional Customer


CUSTOMER_NAME                          NUMBER_OF_ORDERS TOTAL_SPENDING
-------------------------------------- ---------------- --------------
PURCHASE_FREQUENCY
--------------------
Ananya Roy                                            2           2348
Frequent Customer

Sneha Gupta                                           1            450
Occasional Customer

Ishita Menon                                          1            799
Occasional Customer


CUSTOMER_NAME                          NUMBER_OF_ORDERS TOTAL_SPENDING
-------------------------------------- ---------------- --------------
PURCHASE_FREQUENCY
--------------------
Rohan Verma                                           1            999
Occasional Customer

Vikram Kapoor                                         1            899
Occasional Customer

Kabir Singh                                           1            499
Occasional Customer
```

```
CUSTOMER_NAME                          NUMBER_OF_ORDERS TOTAL_SPENDING
-------------------------------------- ---------------- --------------
PURCHASE_FREQUENCY
--------------------
Aditya Mishra                                         1            699
Occasional Customer

Rohan Mehta                                           1            599
Occasional Customer

Nikhil Bose                                           1            550
Occasional Customer


CUSTOMER_NAME                          NUMBER_OF_ORDERS TOTAL_SPENDING
-------------------------------------- ---------------- --------------
PURCHASE_FREQUENCY
--------------------
Kavya Das                                             1            750
Occasional Customer

Meera Rao                                             1            650
Occasional Customer

Anika Iyer                                            1            850
Occasional Customer


CUSTOMER_NAME                          NUMBER_OF_ORDERS TOTAL_SPENDING
-------------------------------------- ---------------- --------------
PURCHASE_FREQUENCY
--------------------
Pooja Reddy                                           1            799
Occasional Customer

Diya Patel                                            1            599
Occasional Customer

Rahul Joseph                                          1            399
Occasional Customer


CUSTOMER_NAME                          NUMBER_OF_ORDERS TOTAL_SPENDING
-------------------------------------- ---------------- --------------
PURCHASE_FREQUENCY
--------------------
Neha Varma                                            1            550
Occasional Customer


19 rows selected.

SQL>
```

7.3 Report 3: Product Performance Report (BOOKS)

- **Product Name (Book Title)**
- **Quantity Sold**
- **Revenue Generated**

```
SQL> SELECT B.TITLE AS PRODUCT_NAME,
  2  SUM(OD.QUANTITY) AS QUANTITY_SOLD,
  3  SUM(OD.QUANTITY * OD.PRICE) AS REVENUE_GENERATED
  4  FROM BOOK B
  5  JOIN ORDER_DETAILS OD
  6  ON B.BOOK_ID = OD.PRODUCT_ID
  7  GROUP BY B.TITLE
  8  ORDER BY REVENUE_GENERATED DESC;

PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------
Designing Data-Intensive Applications
            1              1250

Dune
            2              1198

Harry Potter and the Philosopher's Stone
            1               999


PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------
The Great Gatsby
            2               998

Wings of Fire
            1               899

The Hobbit
            1               850


PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------
The Shining
            1               799

Gone Girl
            1               799

Angels and Demons
            1               750


PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------
The Silent Patient
            1               699

Percy Jackson and the Lightning Thief
            1               699

Foundation
            1               650
```

```
PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------
Me Before You
            1               650

Murder on the Orient Express
            1               599

Dracula
            1               550


PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------
Around the World in 80 Days
            1               550

Sherlock Holmes
            1               499

The Diary of a Young Girl
            1               450


PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------
Pride and Prejudice
            1               450

Treasure Island
            1               399


20 rows selected.
```

7.4 Report 4: Category Analysis Report (BOOK CATEGORIES)

- **Category Name (Book Category)**
- **Total Products Sold (Books)**
- **Total Revenue**

```
SQL> SELECT C.CATEGORY_NAME,
  2  SUM(OD.QUANTITY) AS TOTAL_PRODUCTS_SOLD,
  3  SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  4  FROM CATEGORY C
  5  JOIN BOOK B
  6  ON C.CATEGORY_ID = B.CATEGORY_ID
  7  JOIN ORDER_DETAILS OD
  8  ON B.BOOK_ID = OD.PRODUCT_ID
  9  GROUP BY C.CATEGORY_NAME;

CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD TOTAL_REVENUE
------------------- -------------
Fiction
                  2           998

Sci-Fi
                  3          1848

Mystery
                  1           499


CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD TOTAL_REVENUE
------------------- -------------
Biography
                  1           899

Young Adult
                  1           699

Technology
                  1          1250


CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD TOTAL_REVENUE
------------------- -------------
Thriller
                  2          1498

Romance
                  2          1100

Horror
                  2          1349


CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD TOTAL_REVENUE
------------------- -------------
Adventure
                  2           949

Fantasy
                  2          1849

Crime
                  2          1349
```

```
CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD TOTAL_REVENUE
------------------- -------------
History
                  1           450


13 rows selected.
```

CONCLUSION

The Sales and Customer Analytics System successfully applies SQL aggregate functions to evaluate customer purchasing behavior, sales performance, product performance (Books), and category-wise revenue trends (Book Categories). The generated reports help management identify top customers, best-selling products (Books), profitable categories (Book Categories), and overall bookstore performance.